# Vectors I - HW1: Introduction to Vectors in Three Dimensions

**Solutions**

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Original practice questions. Total: 50 marks. Give exact answers unless otherwise stated.

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**Question 1** **[5 marks]**

The vectors $\mathbf{a}$ and $\mathbf{b}$ are given by $\mathbf{a}=2\mathbf{i}-\mathbf{j}+3\mathbf{k}$ and $\mathbf{b}=-\mathbf{i}+4\mathbf{j}$.

(a) Write $\mathbf{a}$ and $\mathbf{b}$ as column vectors and find $\mathbf{a}+2\mathbf{b}$. [2]

(b) Find $\mathbf{c}$ such that $2\mathbf{a}-3\mathbf{b}+\mathbf{c}=\mathbf{0}$. [2]

(c) Points $P$ and $Q$ have position vectors $\mathbf{a}+2\mathbf{b}$ and $\mathbf{c}$ respectively. Find $\overrightarrow{PQ}$. [1]

**Solution**

(a) $\mathbf{a}=\begin{pmatrix}2\\-1\\3\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}-1\\4\\0\end{pmatrix}$, $\mathbf{a}+2\mathbf{b}=\begin{pmatrix}2\\-1\\3\end{pmatrix}+2\begin{pmatrix}-1\\4\\0\end{pmatrix}=\begin{pmatrix}0\\7\\3\end{pmatrix}$.

(b) $\mathbf{c}=3\mathbf{b}-2\mathbf{a}=\begin{pmatrix}-3\\12\\0\end{pmatrix}-\begin{pmatrix}4\\-2\\6\end{pmatrix}=\begin{pmatrix}-7\\14\\-6\end{pmatrix}$.

(c) $\overrightarrow{PQ}=\overrightarrow{OQ}-\overrightarrow{OP}=\begin{pmatrix}-7\\14\\-6\end{pmatrix}-\begin{pmatrix}0\\7\\3\end{pmatrix}=\begin{pmatrix}-7\\7\\-9\end{pmatrix}$.

**Question 2** **[6 marks]**

The points $A$ and $B$ have coordinates $(1,-2,2)$ and $(3,1,-4)$ respectively.

(a) Find a unit vector that is parallel to $AB$. [2]

(b) Find a vector of length 21 parallel to $AB$. [2]

(c) A point $P$ is such that $\overrightarrow{AP}$ has magnitude 14 and the same direction as $\overrightarrow{AB}$. Find the coordinates of $P$ and state the order of the three points along their common straight path. [2]

**Solution**

(a)
$$\overrightarrow{AB}=\begin{pmatrix}3\\1\\-4\end{pmatrix}-\begin{pmatrix}1\\-2\\2\end{pmatrix}=\begin{pmatrix}2\\3\\-6\end{pmatrix}, \qquad |\overrightarrow{AB}|=\sqrt{4+9+36}=7.$$

One such unit vector is $\frac{1}{7}\begin{pmatrix}2\\3\\-6\end{pmatrix}$. Its negative is also a valid answer.

(b) One such vector is $21\left(\frac{1}{7}\begin{pmatrix}2\\3\\-6\end{pmatrix}\right)=\boxed{\begin{pmatrix}6\\9\\-18\end{pmatrix}}$. Its negative is also a valid answer.

(c)
$$\overrightarrow{AP}=14\left(\frac{1}{7}\begin{pmatrix}2\\3\\-6\end{pmatrix}\right)=\begin{pmatrix}4\\6\\-12\end{pmatrix}, \qquad \overrightarrow{OP}=\overrightarrow{OA}+\overrightarrow{AP}=\begin{pmatrix}5\\4\\-10\end{pmatrix}.$$

Thus $\boxed{P(5,4,-10)}$. Since $\overrightarrow{AP}=2\overrightarrow{AB}$, the order is $A, B, P$.

**Question 3** **[6 marks]**

The vectors $\mathbf{p}=\begin{pmatrix}6\\k\\-3\end{pmatrix}$ and $\mathbf{q}=\begin{pmatrix}2\\-4\\m\end{pmatrix}$ are parallel, where $k$ and $m$ are real constants.

(a) Find $k$ and $m$. [3]

(b) State whether $\mathbf{p}$ and $\mathbf{q}$ have the same or opposite directions, and find $|\mathbf{p}|:|\mathbf{q}|$. [1]

(c) Find all vectors of magnitude $\sqrt{21}$ that are parallel to $\mathbf{p}$. [2]

**Solution**

(a) Since the vectors are parallel, $\mathbf{p}=h\mathbf{q}$ for some non-zero real $h$. Comparing first components gives $6=2h$, so $h=3$.

$$k=3(-4)=-12, \qquad -3=3m \quad \Longrightarrow \quad \boxed{k=-12,\ m=-1}.$$

(b) Since $\mathbf{p}=3\mathbf{q}$, they have the same direction and $\boxed{|\mathbf{p}|:|\mathbf{q}|=3:1}$.

(c) $|\mathbf{q}|=\sqrt{2^2+(-4)^2+(-1)^2}=\sqrt{21}$. Both directions must be included, so all such vectors are

$$\boxed{\pm\begin{pmatrix}2\\-4\\-1\end{pmatrix}}.$$

**Question 4** **[6 marks]**

The points $A$, $B$ and $C$ have coordinates $(1,2,-1)$, $(3,1,2)$ and $(7,-1,8)$ respectively.

(a) Show that $A$, $B$ and $C$ are collinear, and find $AB:BC$. [3]

(b) The point $R(u,-2,v)$ lies on the same straight path as $A$, $B$ and $C$. Find $u$ and $v$. [3]

**Solution**

(a)

$$\overrightarrow{AB}=\begin{pmatrix}2\\-1\\3\end{pmatrix}, \qquad \overrightarrow{BC}=\begin{pmatrix}4\\-2\\6\end{pmatrix}=2\overrightarrow{AB}.$$

Thus $\overrightarrow{AB}$ and $\overrightarrow{BC}$ are parallel and have the common point $B$, so $A, B, C$ are collinear. Taking magnitudes gives $\boxed{AB:BC=1:2}$.

(b) Collinearity gives $\overrightarrow{AR}=s\overrightarrow{AB}$ for some real $s$, so

$$\begin{pmatrix}u-1\\-4\\v+1\end{pmatrix}=s\begin{pmatrix}2\\-1\\3\end{pmatrix}.$$

The second components give $s=4$. Hence $u-1=8$ and $v+1=12$, giving $\boxed{u=9,\ v=11}$.

**Question 5** **[8 marks]**

The points $A$ and $B$ have coordinates $(2,-1,3)$ and $(-4,5,0)$ respectively. A point $M$ divides $AB$ internally in the ratio $AM:MB=2:1$. A point $N$ lies on $AB$ produced beyond $B$ and satisfies $AN:NB=5:2$.

(a) Find the position vector of $M$. [2]

(b) Find the position vector of $N$. [3]

(c) State the order of the four points $A$, $B$, $M$ and $N$, and find $MB:BN$. [3]

**Solution**

(a) By the ratio theorem,

$$\overrightarrow{OM}=\frac{\overrightarrow{OA}+2\overrightarrow{OB}}{3}=\frac{1}{3}\left[\begin{pmatrix}2\\-1\\3\end{pmatrix}+2\begin{pmatrix}-4\\5\\0\end{pmatrix}\right]=\boxed{\begin{pmatrix}-2\\3\\1\end{pmatrix}}.$$

(b) Since $AN=AB+BN$ and $AN:BN=5:2$, $AB:BN=3:2$. Hence $\overrightarrow{BN}=\frac{2}{3}\overrightarrow{AB}$, and

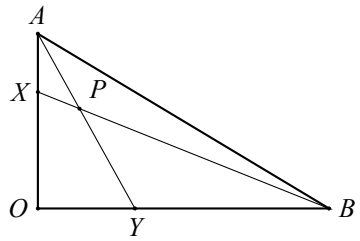
$$\overrightarrow{ON}=\overrightarrow{OB}+\frac{2}{3}(\overrightarrow{OB}-\overrightarrow{OA})=\frac{5\overrightarrow{OB}-2\overrightarrow{OA}}{3}=\boxed{\begin{pmatrix}-8\\9\\-2\end{pmatrix}}.$$

(c) The order is $A, M, B, N$. Since $MB=\frac{1}{3}AB$ and $BN=\frac{2}{3}AB$,

$$\boxed{MB:BN=1:2}.$$

**Question 6** **[10 marks]**

Relative to the origin $O$, the points $A$ and $B$ have non-zero, non-parallel position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. The point $X$ lies on $OA$ with $OX:XA=2:1$, and the point $Y$ lies on $OB$ with $OY:YB=1:2$. The line segments $AY$ and $BX$ intersect at $P$.




(a) By expressing $\overrightarrow{OP}$ in two ways using the ratio theorem, find $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [6]

(b) Find $AP:PY$ and $BP:PX$. [2]

(c) The point $Z$ divides $AB$ internally with $AZ:ZB=1:4$. Show that $O, P, Z$ are collinear and find $OP:PZ$. [2]

**Solution**

(a) First, $\overrightarrow{OX}=\frac{2}{3}\mathbf{a}$ and $\overrightarrow{OY}=\frac{1}{3}\mathbf{b}$. Let $AP:PY=s:(1-s)$ and $BP:PX=t:(1-t)$, where $0<s,t<1$. By the ratio theorem,

$$\overrightarrow{OP}=(1-s)\overrightarrow{OA}+s\overrightarrow{OY}=(1-s)\mathbf{a}+\frac{s}{3}\mathbf{b},$$
$$\overrightarrow{OP}=(1-t)\overrightarrow{OB}+t\overrightarrow{OX}=\frac{2t}{3}\mathbf{a}+(1-t)\mathbf{b}.$$

Since $\mathbf{a}$ and $\mathbf{b}$ are non-parallel, comparing coefficients gives

$$1-s=\frac{2t}{3}, \qquad \frac{s}{3}=1-t.$$

The second equation gives $s=3-3t$. Substituting into the first,

$$-2+3t=\frac{2t}{3} \quad\Longrightarrow\quad 7t=6 \quad\Longrightarrow\quad t=\frac{6}{7}, \quad s=\frac{3}{7}.$$

Consequently,

$$\boxed{\overrightarrow{OP}=\frac{4}{7}\mathbf{a}+\frac{1}{7}\mathbf{b}}.$$

(b) $\boxed{AP:PY=3:4, \qquad BP:PX=6:1}$.

(c) By the ratio theorem,

$$\overrightarrow{OZ}=\frac{4\mathbf{a}+\mathbf{b}}{5}, \qquad \overrightarrow{OP}=\frac{4\mathbf{a}+\mathbf{b}}{7}=\frac{5}{7}\overrightarrow{OZ}.$$

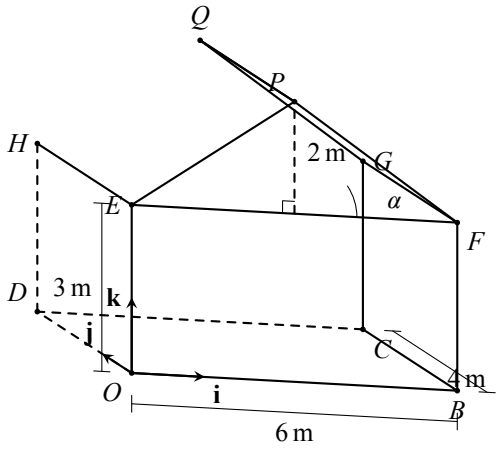
The two vectors are parallel and have the common point $O$, so $O, P, Z$ are collinear. Since $0<\frac{5}{7}<1$, $P$ is between $O$ and $Z$; hence $\boxed{OP:PZ=5:2}$.

**Question 7** **[6 marks]**

A solid house consists of a cuboid $OBCDEFGH$ and a triangular prism roof $EFPHGQ$, as shown. The base $OBCD$ is horizontal. The vertices $E$, $F$, $G$ and $H$ are directly above $O$, $B$, $C$ and $D$ respectively.

The front-left corner $O$ is the origin. The positive **i**, **j** and **k** directions are along $OB$, $OD$ and $OE$ respectively, and each unit represents 1 metre. It is given that $OB = 6$ m, $OD = 4$ m and $OE = 3$ m.

The ridge $PQ$ is parallel to $OD$. The front gable $EFP$ and back gable $HGQ$ are congruent isosceles triangles, with $P$ vertically above the midpoint of $EF$ and $Q$ vertically above the midpoint of $HG$. Each sloping roof face makes an angle $\alpha$ with the horizontal, where $\tan\alpha = \frac{2}{3}$; the vertical roof rise is 2 m.




All lengths are in metres. The diagram is not drawn to scale.

Find the coordinates of all the vertices of the house $(O, B, C, D, \ldots)$.

**Solution**

The positive coordinate directions are along $OB$, $OD$ and $OE$. Hence the four base vertices have $z = 0$, the front vertices have $y = 0$, and the back vertices have $y = 4$.

The walls are 3 m high, so $E, F, G, H$ all have $z = 3$. The roof is symmetric across the middle of its 6 m width, so the ridge has $x = \frac{6}{2} = 3$.

Writing $h$ for the vertical roof rise,

$$\tan\alpha = \frac{h}{6/2} = \frac{2}{3} \quad \Rightarrow \quad h = 2\text{ m}.$$

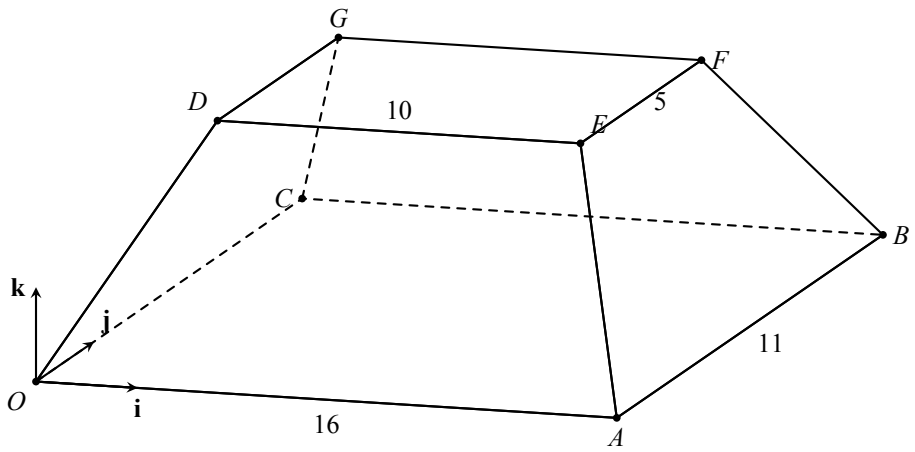
This agrees with the labelled rise. The ridge is at height $3 + 2 = 5$ m. Therefore all ten vertices are

$$\begin{array}{|ll|}\hline O=(0,0,0), & B=(6,0,0),\\ C=(6,4,0), & D=(0,4,0),\\ E=(0,0,3), & F=(6,0,3),\\ G=(6,4,3), & H=(0,4,3),\\ P=(3,0,5), & Q=(3,4,5).\\ \hline\end{array}$$

**Question 8** **[3 marks]**

The diagram shows a container with a horizontal rectangular base $OABC$, where $OA = 16$ cm and $AB = 11$ cm. The top of the container $DEFG$ is also a horizontal rectangle, where $DE = 10$ cm and $EF = 5$ cm. The four sloping faces (for example $OAED$) of the container, each a trapezium, are inclined at the same angle to the horizontal, and the vertical distance between the base and the top is 6 cm.

The point $O$ is taken as the origin $(0, 0, 0)$, and perpendicular unit vectors **i**, **j**, **k** are such that **i** is parallel to $OA$, **j** is parallel to $OC$, and **k** is vertical.




All lengths are in centimetres. The diagram is not drawn to scale.

Find the coordinates of $D$.

**Solution**

Since the four sloping faces are inclined at the same angle to the horizontal, the top rectangle $DEFG$ is centred above the base rectangle $OABC$.

The horizontal inset from each base edge to the corresponding top edge is

$$\frac{OA - DE}{2} = \frac{16 - 10}{2} = 3, \qquad \frac{AB - EF}{2} = \frac{11 - 5}{2} = 3.$$

The top lies 6 cm above the base, so the $z$-coordinate of every top vertex is 6. Taking $D$ as the front-left vertex of the top,

$$\boxed{D = (3, 3, 6)}.$$

**END OF HOMEWORK**